MULT20015 Elements of Quantum Computing
Assignment 1

Due: 2024-08-22 5pm (AEST)
Worth: Total points: 20. Weight for subject mark: 20%.
Number of problems: 4.

Submit your completed written work electronically as a PDF (no other formats accepted) to Canvas/LMS. You are welcome to use an app like `CamScanner` to convert your handwritten notes to pdf, and keep submissions under 10MB. Please put your name and student number on the front of your submission. Show all working.

While we encourage you to discuss this assignment with your classmates (*e.g.*, you can organize informal discussion session or meetups), every student must write their own solution based on their personal understanding. Copying full or partial solutions from other students may invalidate your submission.

1 Single Qubit States [0.5+1.5+1.0 = 3.0 points]

This question aims to solidify our knowledge of single qubit states. In class, we said that the data or "quantum state" of any single qubit register can be expressed as $|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$ where a_0, a_1 are complex numbers.

(a) Consider the state $|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle$ where $a_0 = \sqrt{204/239}$ and $a_1 = i\sqrt{35/239}$.
Plot the amplitudes a_0 and a_1 in the complex plane. [Hint: draw both the real and imaginary axis and mark the point co-ordinates.]

(b) Convert the state to polar notation form: $|\psi\rangle = |a_0|e^{i\theta_0} |0\rangle + |a_1|e^{i\theta_1} |1\rangle$with the basis state phase angles θ_0 and θ_1 in **radians** (expressed as multiples of π).

(c) Now convert the state to "Bloch sphere" form $|\psi\rangle = e^{i\theta_{\text{global}}} \left(\cos\frac{\theta_B}{2} |0\rangle + e^{i\phi_B} \sin\frac{\theta_B}{2} |1\rangle\right)$, specifying the global phase θ_{global} and the Bloch angles θ_B and ϕ_B. Then plot the state on the Bloch sphere.

2 Single Qubit Gates [1.5+3.0+2.0 = 6.5 points]

This question aims to solidify our knowledge of single qubit gates. In class, we said that any single qubit logic gate (or single qubit program) can be represented as a 2×2 matrix.

(a) Consider the following single-qubit operators in matrix form, corresponding to rotations by angle θ_R about some axis with the global phase θ_{global} explicitly included

$$R_X(\theta_R) = e^{i\theta_{\text{global}}} \begin{bmatrix} \cos\frac{\theta_R}{2} & -i\sin\frac{\theta_R}{2} \\ -i\sin\frac{\theta_R}{2} & \cos\frac{\theta_R}{2} \end{bmatrix}$$

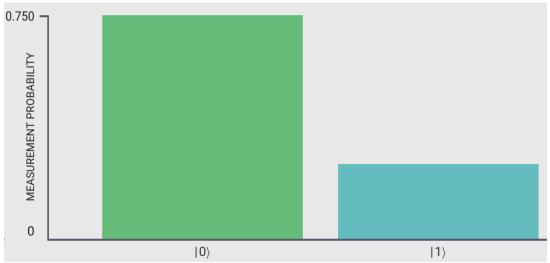
$$R_{X+Z}(\theta_R) = e^{i\theta_{\text{global}}} \begin{bmatrix} \cos\frac{\theta_R}{2} - \frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} & -\frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} \\ -\frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} & \cos\frac{\theta_R}{2} + \frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} \end{bmatrix}$$

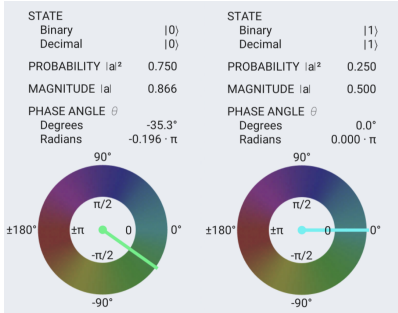
Both gates are π rotations around some axis. The first gate is a rotation about the X axis and corresponds to the gate $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. The second gate corresponds to a rotation about the axis in between the X and Z axis. We call this gate the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Find the value of the global phase for both gates so that $R_X(\pi) = X$ and $R_{X+Z}(\pi) = H$.

(b) In matrix form a rotation by θ_R about the Y+Z axis, ie. $R_{Y+Z}(\theta_R)$, is given by the 2×2 matrix

$$R_{Y+Z}(\theta_R) = \begin{bmatrix} \cos\frac{\theta_R}{2} - \frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} & -\frac{1}{\sqrt{2}}\sin\frac{\theta_R}{2} \\ \frac{1}{\sqrt{2}}\sin\frac{\theta_R}{2} & \cos\frac{\theta_R}{2} + \frac{i}{\sqrt{2}}\sin\frac{\theta_R}{2} \end{bmatrix}$$

Where we have set the global phase θ_{global} to zero. This can be programmed into the QUI with a particular angle θ_R to produce the following state from the initial state $|0\rangle$:





Find the angle θ_R that produced the state $|\psi\rangle$, in both radians (multiples of π) and degrees. Show your working. Explain qualitatively (using diagrams) how the operation moves the state around the Bloch sphere.

(c) Consider a general single qubit program U. In matrix notation, we are given U parameterised by $\alpha, \beta \in \mathbb{C}$:

$$U = \begin{bmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{bmatrix}$$

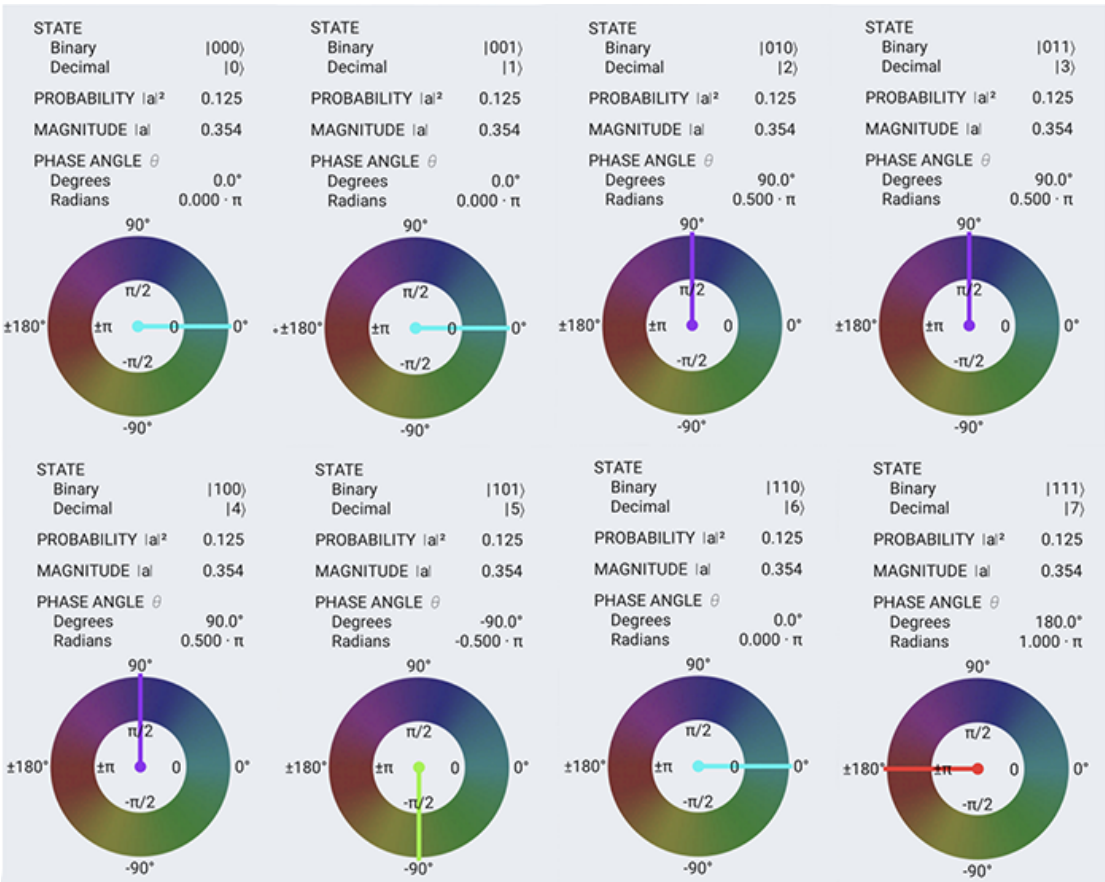
where α^* denotes the complex conjugate of α. It is easy to show that U is unitary, i.e. $U^\dagger U = UU^\dagger = I$ where $\dagger$ denotes transpose then complex conjugate and I is the 2×2 identity matrix, if $|\alpha|^2 + |\beta|^2 = 1$. For the matrices below explain why they are not unitaries and therefore not valid quantum programs

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad M_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} i & 1 \\ 1 & -1 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

3 Multi-qubit states [0.5+2.0+1.0 = 3.5 points]

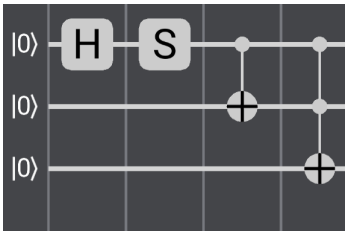
This question aims to solidify our knowledge of multiqubit states.

(a) Consider a three-qubit state for which the QUI state information cards are:



Write out the three-qubit state in ket form in a decimal representation with phases in polar notation

(b) Consider the following circuit over 3 qubits:

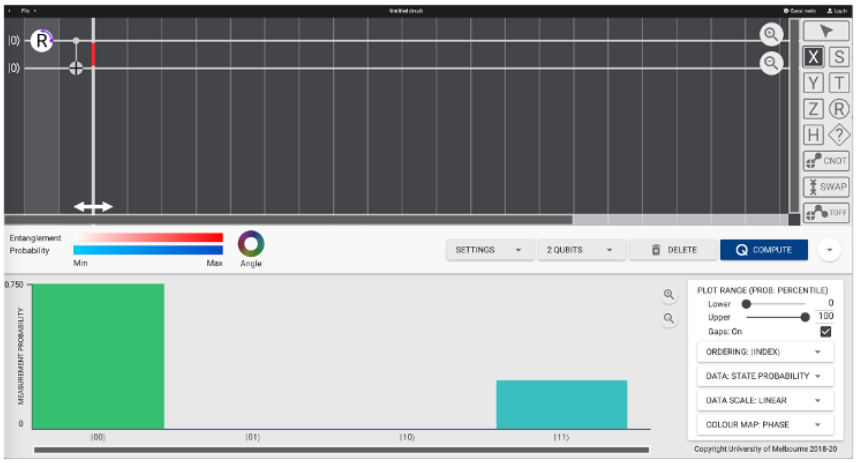


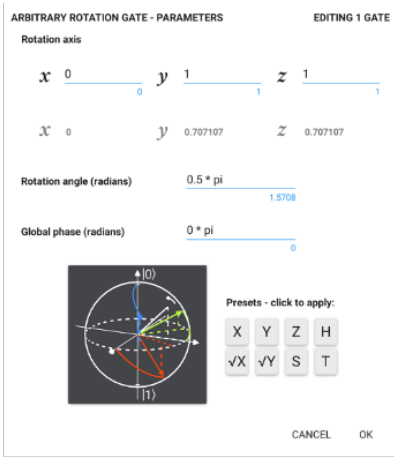
At each time step, write the state in ket form and show your working.

(c) For the state you have calculated in part (b) compute the probability of measuring each of the basis states. That is compute the probability of measuring any of the eight basis states $|000\rangle, |001\rangle, |010\rangle, ..., |111\rangle$ e.g. $\Pr(000) = ?$.

4 Entanglement [0.5+(0.5+2.0+0.5)+2.0+1.5 = 7.0 points]

This question aims to solidify our knowledge of entanglement in multiqubit systems. The following screenshot form QUI will be useful for the below question.





(a) From the information provided, reproduce this program in the QUI to determine the final state produced in the ket form, listing the amplitudes

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|1\rangle \ . \tag{1}$$

(b) The amount of entanglement entropy in the state $|\psi\rangle$ is reported in the QUI by hovering over the time slider between qubit lines – in this case it is 0.811 bits. From your answer in (a), calculate the entanglement entropy between two qubits by the following procedure.

i Construct the matrix

$$A = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} .$$

ii Calculate the eigenvalues λ_1 and λ_2 of the matrix product $AA^\dagger$ where $A^\dagger$ is the Hermitian conjugate of the matrix A.

iii The engagement entropy between the qubits (in units of bits) is given by:

$$H = -\lambda_1 \log_2(\lambda_1) - \lambda_2 \log_2(\lambda_2) \, . \tag{2}$$

For the state in question, $|\psi\rangle$, find the eigenvalues λ_1 and λ_2 and calculate the entanglement entropy between the two quibts in units of bits. Confirm that this value matches that reported by the QUI.

(c) Does global phase affect the entanglement value? Justify your answer.

(d) odify the rotation angle of the R-gate in the initial circuit such that the entanglement entropy is halved. Find the angle in R-gate that halves the entanglement entropy.